

PHASE 2

REQUIREMENT ANALYSIS

2.1 UNDERSTANDING BUSINESS REQUIREMENTS

The first phase of the project focused on understanding the real-world challenges faced by property managers and tenants in managing lease agreements. Through stakeholder meetings, user interviews, and domain research, key pain points were identified to shape the system's core functionalities.

2.2 IDENTIFIED PROBLEMS & USER NEEDS

- Manual handling of lease documents often leads to data entry errors, delays, and lack of consistency.
- Property managers face difficulties in tracking rent payments, lease expirations, and due dates across multiple tenants.
- Lease data is scattered and unstructured, making it hard to access and manage centrally.
- No alert mechanism for upcoming lease renewals or terminations causes missed actions and delays.
- Current systems lack data security, putting sensitive tenant and financial data at risk.

2.3 SOLUTION APPROACH

The Lease Management System addresses these problems by offering an automated, centralized, and secure digital platform. The system is designed to streamline daily operations, enhance visibility, and improve accuracy in lease tracking and management.

2.4 DEFINING PROJECT SCOPE AND OBJECTIVES

This step involves outlining what Phase 1 of the Lease Management System will deliver and how it will align with the overall business goals. The focus is on building a core functional system that can later be expanded.

- Centralized platform for managing leases, tenant records, properties, and payments

- User access through secure login based on roles (admin, manager, tenant)
- Lease lifecycle automation: creation, renewal reminders, and termination alerts
- Dashboard displaying lease status, payment due dates, and tenant activity
- Real-time synchronization of updates to ensure consistent and current data
- Flexibility to scale across multiple properties, locations, and user roles
- Data security measures like authentication, access control, and audit trails

2.5 BUILT-IN REPORTING FOR FINANCIAL SUMMARIES AND LEASE PERFORMANCE OBJECTIVES

- Improve operational efficiency by digitizing manual lease management
- Provide transparency and real-time visibility into lease data
- Enhance communication between tenants and property managers
- Support better decision-making through reporting and analytics
- Ensure system stability and ease of use for non-technical users
- Maintain long-term sustainability with modular and scalable design

The objective is to deliver a reliable MVP (Minimum Viable Product) that solves immediate pain points and forms the foundation for future enhancements.

2.6 DATA MODEL AND SECURITY MODEL

A. DATA MODEL DESIGN

The data model is the backbone of the Lease Management System. It defines how data is structured, stored, and linked across the application. The model ensures efficient data retrieval and clear relationships between different business entities.

Core Entities:

- **Users:** Stores login credentials and role information (Admin, Property Manager, Tenant)
- **Tenants:** Contains personal information, lease history, contact details, and tenant-specific notes
- **Properties:** Stores property details like ID, name, address, and ownership information
- **Leases:** Connects tenants to properties and includes start/end dates, rent amount, and lease status

- **Payments:** Tracks payment records including amount, due date, payment status, method, and remarks Relationships:

- One Property → Many Leases
- One Tenant → Many Leases (historical)
- One Lease → Many Payments

These relationships ensure a normalized, scalable data structure that supports reporting and performance optimization.

B. SECURITY MODEL DESIGN

Given the sensitivity of lease and financial data, the system incorporates strong security practices to safeguard information and ensure data privacy.

- **User Authentication:**

A secure login system requiring valid username and password combinations, with session control to prevent unauthorized access.

- **Role-Based Access Control (RBAC):**

Users are assigned roles which define what data they can view or modify.

- Tenants can view their own leases and payments.
- Property Managers can manage leases, payments, and tenant profiles.
- Admins have full access to all modules.

- **Data Encryption:**

Sensitive information (e.g., payment methods, user credentials) is encrypted both during transmission (SSL/TLS) and at rest in the database.

- **Audit Logging:**

Every critical operation (e.g., lease updates, payment status changes) is recorded for traceability. This ensures accountability and helps in debugging or compliance.

- **Input Validation & Protection:**

All user input is validated to prevent common web vulnerabilities such as:

- SQL Injection
- CrossSite Scripting (XSS)
- CSRF (Cross-Site Request Forgery)

These security layers protect the system from unauthorized access, data loss, and malicious attacks, ensuring the platform is trustworthy and compliant with best practices.